direction at a constant angular velocity ω. Such a rotating vector is called a phasor. A phasor is drawn in such a way that

- the length of the line segment equals the peak value V_m (or I_m) of the alternating voltage (or current)
- its angular velocity ω is equal to the angular frequency of the alternating voltage (or current)
- the projection of phasor on any vertical axis gives the instantaneous value of the alternating voltage (or current)
- the angle between the phasor and the axis of reference (positive x-axis) indicates the phase of the alternating voltage (or current).

The notion of phasors is introduced to analyse phase relationship between voltage and current in different AC circuits.

Phasor diagram

The diagram which shows various phasors and their phase relations is called phasor diagram. Consider a sinusoidal alternating voltage $v = V_m \sin \omega t$ applied to a circuit. This voltage can be represented by a phasor, namely $\overrightarrow{OA}$ as shown in Figure 4.43.

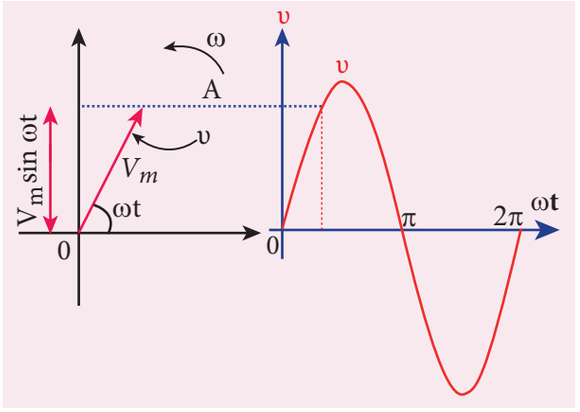


Figure 4.43 Phasor diagram for an alternating voltage $v = V_m \sin \omega t$

Here the length of $\overrightarrow{OA}$ equals the peak value (V_m), the angle it makes with x-axis gives the phase (ωt) of the applied voltage. Its projection on y-axis provides the instantaneous value ($V_m \sin \omega t$) at that instant.

When $\overrightarrow{OA}$ rotates about O with angular velocity ω in anti-clockwise direction, the

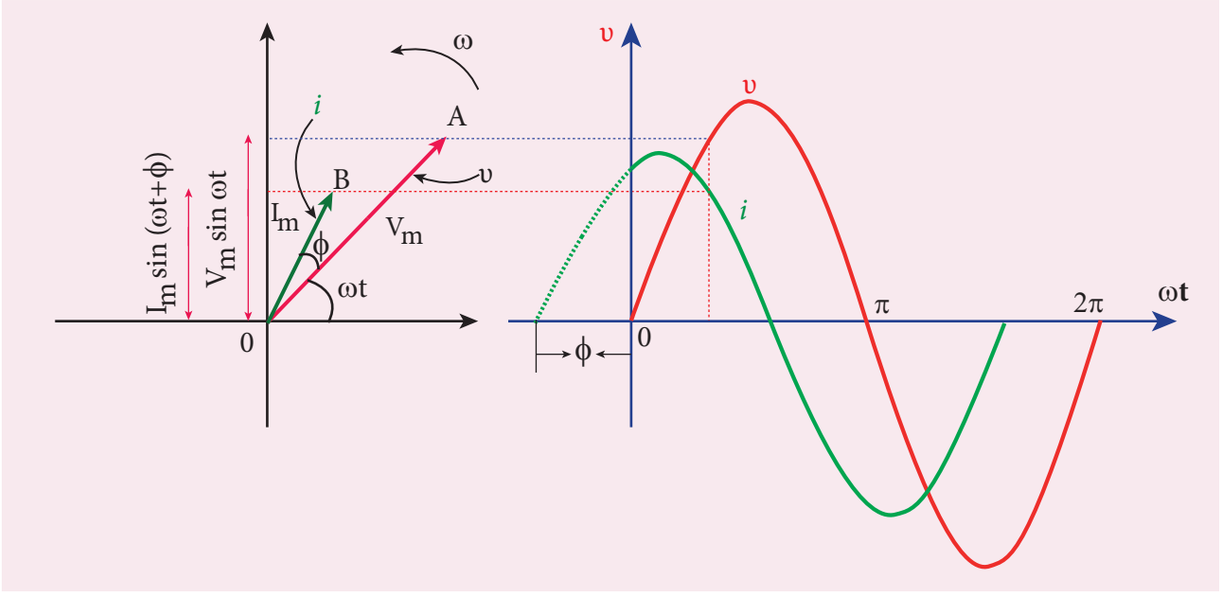


Figure 4.44 Phasor diagram and wave diagram say that i leads v by ϕ

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